



POKHARA UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2026 Fall		
Level	B. E.	F M	100
Programme	Bachelor	PM	45
Year/Part	2 nd Year (3 rd Semester)	Time	3 hours

Candidates are required to give their answer in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all questions.

1. a) Evaluate the integral: $\int_0^{a/\sqrt{2}} \int_y^{\sqrt{a^2-y^2}} \log(x^2 + y^2) dx dy$. 5
- b) Find the volume of the solid whose base in the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. 5
- c) Evaluate the integral: $\int_0^{\log 2} \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx$. 5
2. (a) (i) Express $4x^3 + 6x^2 + 7x + 2$ in terms of Legendre polynomial. 4+4
- (ii) Show that $(n+1)P_n(x) = P'_{n+1}(x) - xP'_n(x)$.

?

OR

Define Bessel's differential equation of order n . Find its complete solution.

- b) Use the power series to solve the equation $y'' + y = 0$. 7

3. a) (i) Define Laplace transform of a function and find $\mathcal{L}(t \sin^2 t)$. 4+4
- (ii) Find the inverse Laplace transform of the function $\frac{a}{s^2(s^2+a^2)}$

- b) Apply Laplace transform to solve the initial value problem $y'' + 4y' + 3y = e^{-t}, y(0) = y'(0) = 1$. 7

4. a) If $\phi = x^3 + y^3 + z^3 - 3xyz$, find $\text{div}(\text{grad } \phi)$ and $\text{curl}(\text{grad } \phi)$. 5
 b) A particle moves along the curve $(2t^2, t - 4t^2, 3t - 5)$. Find the component of its velocity and acceleration at time $t = 1$ second in the direction $\vec{i} - 3\vec{j} + 2\vec{k}$. 5
 c) Show that $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - xy)\vec{k}$ is irrotational also find the scalar potential. 5

5. a) Evaluate the integral $\int_C (x + y)dx + (2x - z)dy + (y + z)dz$, where C is the boundary of the triangle with vertices $(2,0,0)$, $(0,3,0)$, and $(0,0,6)$ using Stokes' theorem. 8

OR

State Gauss divergence theorem. Use this theorem to evaluate $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = (4x, -2y^2, z^2)$, S is the surface bounded by the region $x^2 + y^2 = 4$ in between the planes $z = 0$ and $z = 3$. 7

- b) State Green's theorem and apply it to evaluate $\oint_C \vec{F} \cdot d\vec{r}$ counterclockwise, where $\vec{F} = (\sin y, \cos x)$ and C is the triangle with vertices $(\pi, 0)$ and $(1, \pi)$.

6. a) Find the Fourier series of the function $f(x) = x^2$ in the interval $[-\pi, \pi]$ and hence show that $\frac{\pi^2}{6} = 1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots$ 7
 b) Find the Fourier sine and cosine series of the function $f(x) = e^x$ in $0 < x < L$. 8

7. Attempt any two questions: $[2 \times 5 = 10]$

- a) Find the general solution of linear first order differential equation $au_x + bu_y + cu = f(x, y)$.
 b) Show that the integral $\int_{(1,0,2)}^{(-2,1,3)} [(6xy^3 + 2z^2) dx + 9x^2y^2 dy + (4xz + 1) dz]$ is exact and evaluate the exact value.
 c) Find the general solution of $u_x + u_y - u = 0$.

*****The End *****

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE Computer (II/I)
Course: Microprocessor and ALP

Internal Exam

Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the historical evolution of Intel microprocessors from early microprocessors to modern processors. Discuss the major architectural improvements introduced during the evolution. 7
b) Define addressing mode. Explain the addressing modes of 8085 microprocessor with suitable examples. 8
2. a) Write an 8085 assembly language program to find the largest number from a series of numbers stored in memory locations C050H–C059H and store the result at C060H. 8
b) Explain instruction cycle, machine cycle, and T-state in 8085 microprocessor. Draw and explain the timing diagram of memory read cycle. 7
3. a) Explain the architecture of 8086 microprocessor with a neat block diagram. Discuss the functions of BIU and EU. 8
b) What is stack? Explain stack operation in 8086 with suitable examples. Differentiate between procedures and macros. 7
4. a) Write an 8086 assembly language program to count the number of characters in a given string. 8
b) Define memory mapped I/O and I/O mapped I/O. Compare them with suitable examples. 7
5. a) What is address decoding? Design an address decoding scheme to interface 8KB RAM and 16KB EPROM with 8085 microprocessor. State whether the decoding is full or partial. 8
b) Explain the architecture and operation of Intel 8255 Programmable Peripheral Interface (PPI). 7
6. a) Explain interrupt processing sequence in 8085 microprocessor. Differentiate between maskable and non-maskable interrupts. 8
b) Explain the handling of multiple interrupts using Intel 8259 Programmable Interrupt Controller with suitable block diagram. 7
7. Write short notes **any two**: 2x
a) Minimum and Maximum mode of 8086
b) Synchronous and Asynchronous data transmission
c) USART (8251) and its applications

POKHARA UNIVERSITY FACULTY OF SCIENCE AND TECHNOLOGY SCHOOL OF ENGINEERING	Exam		Final Internal Examination 2026		
	Level		B.E	F M	100
	Programme		BCE	PM	45
	Year/Part		II/I	Time	3 Hrs
Subject: Computer Graphics					

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicate full marks.
Attempt all the questions

1.	a) Consider a raster scan system having 12-inch by 12-inch screen with a resolution of 100 pixels per inch in each direction. If display controller of this system refresh the screen at the rate of 50 frames per second, how many pixels could be accessed per second and what is the access time per pixel of the system b) Rasterize the line from (0,0) to (6,7) using bresenham line drawing algorithm.	7 8
2.	a) Explain flood fill techniques with its algorithm. b) Window port is given by (100,100,300,300) and viewport is given by (50,50,150,150). Convert the window port coordinate (200,200) to the viewport coordinate.	8 7
3.	a) Perform a 45 degree rotation of triangle A(0,0),B(1,1),C(5,2) about fixed point (-1,-1). b) Use the Cohen -Sutherland algorithm to clip the line P1(70,20) and P2(100,10) against a window lower left hand corner (50,10) and upper right hand corner (80,40)	7 8
4.	a. Describe the rotation of an object about an axis , which is not parallel to any of three coordinate axes of coordinate system. b. Explain Z-buffer algorithm for visible surface determination with suitable example.	8 7
5.	a. Develop an illumination model for a point source considering the effect of ambient light, diffused and specular reflection. b. Why machine independent programming language is used? Discuss any 5 file formats.	7 8
6.	a. Define Aspect ratio and bit depth. How computer graphics is different from image processing? b. Explain beam penetration and shadow mask method for color generation. c. Explain about different techniques for 3D object representation.	5 5 5
7.	Write short notes on (Any Two) a. Back face surface detection method b. CMYK color model c. OpenGL	5 × 2

-- Best of luck --

POKHARA UNIVERSITY
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Level: Bachelor
Programme: Computer III
Course: Operating Systems
Semester: Fall
Year : 2025
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.
Attempt all the questions

2. a) Define OS. Discuss the different functions of operating system.
b) How CPU switches from one process to another? Illustrate with a diagram. What is switching overhead?

OR

Explain Dining Philosopher Problem in process synchronization and give solutions to it.

2.

- a) Consider the following set of processes, with the arrival times and the CPU-burst times given in milliseconds. Calculate the AWT and average TAT using SRTN, FCFS and RR (quantum=2ms).

Process	Arrival Time	Burst Time
A	1	5
B	3	3
C	4	3
D	2	1

- b) Define process. Explain PCB, process states and its transition with figures

3. a) Define page fault. Consider the following page reference strings: 1,3; 8,4,5,3,1,6,7,2,1,3. How many page faults would occur for each of the following page replacement algorithms assuming 4 pages a frame in each case calculate fault ratio.

1

- i. Second Chance page replacement algorithm
ii. LRU page replacement algorithm
iii. Optimal page replacement algorithm
iv. FIFO algorithm
b) Define Relocation and Protection. With suitable diagram explain Inverted Page Table.

OR

- Mention the differences between Paging and Segmentation. What are different page-table structures? Explain any one.
4. a) What is Internal and External Fragmentation. Explain paging with segmentation in detail.

- b) Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time to following snapshot of the system has been taken:

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	0	1	0	7	5	3	3	3	2
P ₁	2	0	0	3	2	2			
P ₂	3	0	2	9	0	2			
P ₃	2	1	1	2	2	2			
P ₄	0	0	2	4	3	3			

- i. What will be the content of need matrix?
ii. Is the system in a safe state? If Yes, then what is the safe sequence?

5. a) What do you understand by design principles in OS? Explain different Input output techniques with diagrams
b) Why we need to organize the file? Explain the ways of file organization with suitable diagrams.

6. a) Portray the characteristics of Cloud Operating System as well as mention the different security issues in OS.

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- b) Define Memory Wall and explain its impacts in OS.
7. Write short notes on **any two**:
- a) Kernel and its Types
 - b) Memory hierarchy
 - c) Bottleneck in OS

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Exam	Final Internal Examination 2025 Fall		
Level	B.E.	F M	100
Program	BoCE	PM	45
Year/ Part	III/ I	Time	3 Hrs

Subject: Data Communication

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks

Attempt all the questions.

1. a) Draw a generic block diagram of digital communication system and briefly explain the function for each block. (4+4)

b) Differentiate between parallel and serial transmission. Briefly explain RS232C interface standards. (7) *- bit parallelly bit flows one by one*

2. a) Explain deterministic and random signal with example. Differentiate between energy and power signal. [4+3] *P, A, F can be determined one can't at any point.*

b) Define linear, stable, time invariant and causal system with examples. (8)

3. a) What is PDU? Differentiate between LLC and MAC sub layer of datalink layer of OSI reference model. (2+6)

b) What is meant by "Open System Interconnection"? Briefly explain the layers of OSI reference model. (7)

OR

Briefly explain the protocols and services offered by TCP/IP layers.

4. a) What do you mean by guided media? Mention the advantages of optical fiber over co-axial cables and twisted pair cables. (2+6)

b) Describe the transmission impairments for communication system with suitable example. (7)

5. a) Define flow control. Briefly explain stop and wait, go-back-N, selective-repeat request ARQ. (2+5)

b) Compare and contrast between circuit switching, packet switching and message switching. (8)

OR

What do you mean by multiplexing? Explain FDM hierarchy in telephone system.

6. a) Differentiate between AM and FM. Why is FM superior over AM in communication? (4+4)

b) What is multilevel modulation? Briefly explain with a suitable example. (7)

7. Write short notes on (Any Two): (2X5)

i) CSMA/CD

ii) Lossy Compression

iii) Bit Rate / Baud Rate

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Exam	Final Internal Examination 2025 Fall		
Level	B.E.	FM	100
Program	BE Computer	PM	45
Year/ Part	2/1st	Time	3 Hrs

Subject: Database Management System

*Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.*

1. a) What do you mean by data independence? How is schema different from instance? Justify with suitable examples. 7
- b) What do you understand by E-R model? Draw an E-R diagram for an assumed business firm Pokhara Fitness Club Pvt. Ltd. (provides facilities for gym, sauna, etc. for members on monthly/annually subscription basis) including primary key, weak entity, composite attribute, derived attribute and multivalued attributes in your ER diagram. 8

2. a) What is a view? Give one example of it. Write importance of data dictionary. 7
- b) Given relation schema as below 8

Employee(emp-id, name, address, telephone, post, salary, age)
Works-on (emp-id, dept-id, join-date.)
Department(dept-id, dept-name, phone, budget)

Write the sql commands for the following.

- i. Insert new record with department id 101, name: Computer, and budget Rs 10 million in Department relation.
- ii. Find the name of the department on which salary of employee is greater than or equal to 20000.
- iii. List the name of employees whose name starts with "A" and ends with "m"
- iv. Find the employee name and department name of those employees who living in address Pokhara.
- v. Increment the salary of the employee by 10% whose post is manager.
- vi. List name of employee whose age is greater than average age of all employees.
- vii. List employee id of all employees who joint project on "06/05/2022"
- viii. List the name of employees whose name starts with N or with K

3. a) Define normalization. Explain about 1NF, 2NF, 3NF and BCNF with suitable examples. 8

- b) What are the roles of Assertions and Triggers in SQL? Explain with examples. 7

4. a) Explain cryptography and its types with related diagram of each. 8

OR

What is NoSQL? What are the types of NoSQL databases? Explain with example.

- b) What is Query optimization? How can it be achieved? 7

5. a) Suppose we are given the following table definitions with the certain records in each table. 8

PROJ (PNO, PNAME, BUDGET)
EMP(ENO, ENAME, TITLE)
ASG(ENO, PNO, DUR)

Write the RA expression for following tasks:

- i) "Find the names of employees other than Shyam Karki who worked on ABC ERP project for either 2 or 3 years".
- ii) Construct **initial operator tree** and final **efficient operator tree** by following the steps in query optimization.

- b) Explain insertion in B+ Tree index file with related example. 7

6. a) Explain conflict and view serializability. Test conflict serializability of the following schedule 8

T1	T2	T3
Read(X)		
	Read(Z)	
Read(Z)		Read(X)
	Read(Y)	
		Read(Y)
Write(X)		
	Write(Z)	
		Write(Y)
	Write(Y)	

b) What are the various types of failures that can occur in database? Discuss the log based recovery mechanism. 7

2x5

7 Write short notes on (any two)

- Denormalization for performance
- Blockchain → ①
- ORM
- Armstrong's Axioms

*** Best of Luck ***